Appendix S1

Combining Temporal, Environmental, Demographic, and Geospatial Information to Forecast Influenza by Neighborhood: A Graph Neural Network in 3 US Cities

Contents. Appendix S1 (models, tuning, data notes, and design experience on the development season); Tables S1–S18; Figures S1–S6.

Models

Table S1 lists every model in the evaluation. All learned models were tuned with the same budget in every test season and averaged over 10 training runs, in all 3 cities; all models except the FluSight baseline received forecast intervals from the same shared procedure described in the main text. Table S2 lists which analyses were run in each city and, where one was not, why.

Analyses performed in each city

Comparison models: ARIMA and LSTM

[Coauthor: describe the ARIMA comparison model here (about 120–170 words).]

[Coauthor: describe the LSTM comparison model here (about 130–180 words).]

Equal-budget tuning

For each learned model, test season, and horizon, 12 configurations were trained once: the model's default and 11 drawn at random from the space in Table S3. The configuration with the lowest mean absolute error on the validation weeks immediately before the test season was retrained with 10 random seeds and scored on the test season. The chosen configurations are recorded in `Code/sweep/tuned_configs.json`.

The weighted interval score

The weighted interval score (WIS) asks how far off a forecast was, counting both its best guess and how honest its range was.[4] A close forecast that claims too much certainty should score worse than an equally close one that admits its uncertainty, and a very wide range should not score well only because it happened to contain the observed value.

The WIS is built from the interval score. For a central $(1-\alpha)\times 100\%$ forecast interval from l to u and an observed value y, the interval score is

$$\mathrm{IS}_\alpha(l,u;y) = (u-l) + \frac{2}{\alpha}(l-y)\,\mathbf{1}(y<l) + \frac{2}{\alpha}(y-u)\,\mathbf{1}(y>u),$$

where $\mathbf{1}(\cdot)$ is 1 when the condition holds and 0 otherwise. The first term is the width of the interval; the other 2 are penalties for missing, proportional to how far outside the interval the observation fell. For a 95% interval ($\alpha = 0.05$), each unit of miss costs 40 times as much as each unit of width, so the score is lowest, on average, for an interval that really does contain the outcome 95% of the time.[5]

Table S1: Models Compared

Model	What it does	Inputs
Graph neural network	Causal temporal convolutions over each neighborhood's recent weeks, graph convolution over the neighborhood map plus a learned set of extra links, and a learned correction to a blend of the current week and a seasonal curve	16 weeks of rates, 6 weather, 8 demographic measures, map
Same model, map removed	As above with no links between neighborhoods and no surrounding areas	Same, no map
Same model, no environment	As the graph neural network without the 6 weather variables	Same, no weather
Same model, no demographics	As the graph neural network without the 8 demographic measures as inputs (demographic links kept)	Same, no demographic inputs
Same model, past rates only	No weather, demographic measures, suppression flag, or links; same baseline	16 weeks of rates
LSTM	Long short-term memory network reading all neighborhoods' series jointly	All series, no map
Cola-GNN	Published influenza graph model that learns connections from the data;[1] reimplemented	All series
Gradient-boosted trees	Regression trees pooled across neighborhoods	Same columns as graph model, no map
Seasonal ARIMA	Per-neighborhood ARIMA with the annual cycle as harmonic regressors	Own history
Seasonal blend, no learning	The graph model's baseline (current week blended with the seasonal curve) with the network switched off	Own history
FluSight baseline	Latest value carried forward, ranges from past changes over the forecast horizon	Own history
Persistence	Next week equals this week	Own history
Climatology	Seasonal curve for the target week	Own history
Same week last year	The value in the same week one year earlier	Own history
Graph attention network[a]	Spatial graph attention over 52 weeks of history, no temporal layer[2,3]	52 weeks, map
Dual-topology GCN[a]	Separate geographic and correlation graphs with temporal convolutions[3]	52 weeks, 2 maps

Abbreviations: ARIMA, autoregressive integrated moving average; GCN, graph convolutional network; LSTM, long short-term memory network.

[a] The comparison and reference models of Luo et al.,[3] run at 1 week ahead only, as in that paper.

Every model in this study reports the 23 quantiles used by FluSight: the median (m) and 11 central intervals with $\alpha_k = 0.02, 0.05, 0.10, 0.20, \ldots, 0.90$ (98%, 95%, 90%, 80%, ..., 10% intervals). The WIS combines them:

$$\mathrm{WIS}(y) = \frac{1}{K + \frac{1}{2}} \left(\tfrac{1}{2} \, |y - m| + \sum_{k=1}^{K} \frac{\alpha_k}{2} \, \mathrm{IS}_{\alpha_k}(l_k, u_k; y) \right), \qquad K = 11.$$

Each interval is weighted by $\alpha_k/2$, so narrow intervals near the median count with small weight and wide intervals with larger weight, and the median alone contributes half its absolute error. The WIS is in the units of the forecast (ED visits for ILI per 100,000, or Chicago's 1–10 levels), and lower is better. A forecast that puts all its quantiles on a single value has a WIS equal to its absolute

Table S2: Analyses Performed in Each City

Analysis	Boston	Columbus	Chicago
Training data used	December 2017 onward	July 2022 onward[a]	September 2019 onward
Unseen test seasons	3 (2022–2023 to 2024–2025)	1 (2024–2025)	4 (2022–2023 to 2025–2026)
Development season (not evidence)	2025–2026	2025–2026	None[b]
Models compared (Table S1)	All 16	All 16	All 16
Equal-budget tuning, 10 training runs	Yes	Yes	Yes
Week-level tests (Tables S7–S9)	Yes	Yes	Yes
Cross-city transfer (main text, Table 5)	Target and source	Target and source	Target and source
Training on several cities at once (Table S18)	Target and source	Target and source	Target and source
Hand-drawn vs automatically built map (main text, Table 4)	Yes	Not applicable[c]	Not applicable[c]
Severity alerts (Tables S13, S14)	Yes	Yes	Not applicable[d]
Component ablations (main text, Table 3, Figure 4; Tables S15, S16)	Yes	Yes	Yes
Map effect by neighborhood; example forecasts and calibration (main text, Figures 5, 6)	Yes	Not shown[e]	Not shown[e]
Design experiments on the development season (Table S17)	Yes	No[f]	No[f]

[a] Influenza nearly disappeared in the first pandemic year, and the Columbus series begins in January 2022, so models start after it; a 2023–2024 test season would have had under 1 year of training data.
[b] Chicago was added after the design was fixed; tuning used only the weeks before each test season.
[c] Only Boston had a hand-drawn map. The Columbus and Chicago maps were built from boundary files from the start, by the rule tested in Table 4.
[d] Chicago's measure is already the health department's graded 1–10 activity level, so there is no rate on which to set intensity thresholds.
[e] Illustrative figures for the primary city; the overall results for the other cities are in Tables S5, S6, S8, and S9, and their component ablations in Table S15.
[f] The model's structure was chosen on Boston's development season only.

error, so WIS can be read on the same scale as mean absolute error. The WIS approximates the continuous ranked probability score, a standard score for whole forecast distributions, and is algebraically equal to twice the mean quantile (pinball) loss over the 23 quantiles, which is how the code computes it. Each table reports the mean WIS over neighborhoods and target weeks; relative WIS divides a model's mean WIS by that of persistence.

The WIS can be split exactly into a spread term (the weighted widths, $\sum_k \frac{\alpha_k}{2}(u_k - l_k)$, divided by $K + \frac{1}{2}$) and penalties for forecasting too high or too low (the weighted misses, including the median term). Most of a wide, cautious forecast's score comes from spread, and most of a narrow forecast's score comes from penalties when it misses.

In Boston, the graph neural network's mean WIS 1 week ahead was 5.13 (Table 2 of the main text): its forecasts were off by about 5 ED visits for ILI per 100,000 residents per neighborhood-week, with credit given for honest ranges. For scale, observed rates in the 3 unseen seasons averaged 26.9 per 100,000 (median, 15.5; 90th percentile, 60.9). The WIS is lower than the same forecasts'

Table S3: Tuning Search Spaces

Model	Values searched
Graph neural network (with and without map)	learning rate {0.001, 0.003, 0.01}; weight decay {0.0003, 0.001, 0.003, 0.01}; dropout {0.1, 0.25, 0.4}; channels {16, 24, 32}; epochs {80, 120, 200}
LSTM	learning rate {0.0003, 0.001, 0.003}; weight decay {0.00001, 0.0001, 0.001}; dropout {0, 0.1, 0.25}; hidden size {32, 64, 128}; lookback {8, 16, 26} weeks; epochs {80, 120, 200}; target {level, blend}
Gradient-boosted trees	learning rate {0.01, 0.03, 0.1}; maximum depth {2, 3, 4, 6}; minimum child weight {1, 4, 8}; row subsample {0.6, 0.8, 1.0}; column subsample {0.6, 0.8, 1.0}; L2 penalty {0.5, 2, 8}; target {climatology, level, blend}
Cola-GNN	learning rate {0.0003, 0.001, 0.003}; weight decay {0, 0.0005, 0.005}; dropout {0.1, 0.2, 0.4}; hidden size {16, 32, 64}; lookback {12, 20, 26} weeks; epochs {100, 150, 200}
Graph attention network	learning rate {0.001, 0.003, 0.01}; weight decay {0.00001, 0.0001, 0.001}; dropout {0.1, 0.2, 0.4}; hidden size {32, 64}; heads {4, 8}; epochs {100, 200}
Dual-topology GCN	learning rate {0.001, 0.003, 0.01}; weight decay {0.00001, 0.0001, 0.001}; epochs {100, 200}; fusion {concatenate, mean, maximum}

Abbreviations: GCN, graph convolutional network; LSTM, long short-term memory network.

mean absolute error (7.85; Table S10) because a forecast whose range covers the outcome is charged less than one that commits to a single value. Of the 5.13, about 1.9 came from the width of the ranges, 2.1 from forecasting too low, and 1.2 from forecasting too high. The mean is driven by high weeks. In the tenth of neighborhood-weeks with the highest rates, the average WIS was 16.2, against 3.9 in the rest, and those weeks supplied about a third of the total; a typical (median) week scored 2.8. WIS values cannot be compared across cities with different measures: Boston's (about 5 to 8) are in rates and Chicago's (about 1 to 1.5) in 1–10 levels. Within a city they also rise and fall with how high a season ran, so models are compared with each other on the same weeks. For this reason the main text leads with relative WIS. Persistence's mean WIS 1 week ahead was 6.24, so the graph neural network's relative WIS was $5.13/6.24 = 0.82$, or 18% less error than persistence.

As an example, consider Dorchester. The graph neural network's forecast made on January 19, 2025, for the week of February 2 (2 weeks ahead) had a median of 226.7 per 100,000 and a 95% interval of 175.6 to 292.6. The observed rate was 294.4, just above the 97.5th percentile. The WIS for this forecast was 43.5, of which 6.1 came from spread and 37.4 from underprediction. The model forecast a rise but too low a peak, and nearly all of the score reflects that shortfall. Persistence, which carried forward the 292.5 observed on January 19, scored 9.0 that week. Forecasts that fall short of peaks, as described in the main text, are penalized most because errors are largest where rates are high.[6]

Data notes by city

Inputs shared by all cities. The 6 weekly weather variables are mean, maximum, and minimum temperature, relative humidity, precipitation, and maximum wind speed, from the Open-Meteo archive. The 8 demographic measures in all 3 cities are population density, poverty, share of children, share aged 65 years or older, homeownership, transit commuting, households without a vehicle, and share of nonwhite residents, from city planning tables in Boston and from American

Community Survey 5-year estimates in Columbus and Chicago. Each measure is scaled within its own city.

Boston. Rates are published weekly by the Boston Public Health Commission for 14 neighborhoods (Back Bay, Beacon Hill, Downtown, North End, and West End form a single neighborhood). Suppressed weeks were treated as missing and excluded from training and scoring. Two data-loading errors found during the project (a count row averaged with the rate row, and suppressed weeks read as zero) were corrected before any result reported here was produced.

Columbus. Visit-level ED records for influenza-like illness from Columbus Public Health were assigned to 17 planning areas through the department's ZIP code crosswalk. Population denominators come from the American Community Survey 5-year estimates summed over ZIP code tabulation areas. Areas with no visits in a week are recorded as observed zeros. The 5 surrounding areas in the map are fringe ZIP codes (43002, 43035, 43140, 43146, and 43240) that the crosswalk lists but leaves unassigned to any planning area; they carry 0.2% of visits.

Chicago. The Chicago Department of Public Health publishes a weekly influenza-like illness activity level for each ZIP code on the City of Chicago Data Portal (dataset 8vvr-jv2g), compiled from Illinois emergency department syndromic surveillance. The level compares a ZIP code's share of emergency department visits for influenza-like illness with that ZIP code's own summer average: level 1 is at or below it, each level above is roughly 1 more standard deviation, and the scale stops at 10, which 13% of ZIP-weeks reach. Because it is an index with no denominator, errors are in levels and cannot be compared in size with the other cities. The series begins in September 2019; the city is absent for May–September 2020, and those weeks are missing. Of 60 ZIP codes, 1 (60106, mostly outside the city) has no boundary and is dropped, and 1 (60827) that joined the feed in September 2020 is in the map but not scored, leaving 58. O'Hare (60666) has no residents and takes the city median for its demographic measures. The map has 115 shared borders, derived from the city's ZIP code boundary file. Models were trained on all weeks from September 2019, and every Chicago season from 2022–2023 to 2025–2026 is unseen. One ZIP code (60707) reported level 1 in every week before June 2022, so in the 2022–2023 training data its series had no variation. Each series is scaled by its training-period standard deviation; a series with none is scaled by 1 instead, which leaves every other series and city unchanged. An earlier run that divided by a near-zero value gave that ZIP code test inputs near 10^9, which graph models passed to its neighbors; all 2022–2023 Chicago results reported here are from the corrected runs.

Full results by season

Table S4: Mean Weighted Interval Score by Model, Season, and Horizon, 14 Neighborhoods, Boston, Massachusetts, 2022–2025

	2022–2023			2023–2024			2024–2025		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	5.56	7.70	10.95	4.70	5.07	5.53	5.14	6.64	8.18
Same model, map removed	6.23	7.83	10.86	5.04	5.37	5.98	5.79	6.95	9.20
LSTM	5.50	7.67	11.29	4.85	5.85	6.84	5.20	6.96	9.33
Cola−GNN	6.01	10.37	10.97	4.79	5.29	6.43	5.24	7.17	8.92
Gradient−boosted trees	6.29	8.09	11.31	5.28	5.61	6.34	6.68	8.27	9.49
Seasonal ARIMA	6.29	8.02	10.94	5.43	5.75	5.91	7.84	9.22	10.60
Seasonal blend, no learning	6.26	7.95	11.30	5.63	6.08	6.44	5.97	7.84	10.96
FluSight baseline	6.65	8.55	12.50	5.83	6.52	7.68	6.69	9.41	14.17
Persistence	6.51	8.30	12.38	5.86	6.57	7.87	6.36	8.95	14.07
Graph attention network[a]	6.32			5.08			5.02		
Dual−topology GCN[a]	11.36			5.38			6.45		
Climatology	13.25	13.25	13.21	6.59	6.60	6.60	11.10	11.10	11.10
Same week last year	12.19	12.19	12.19	13.71	13.79	13.79	10.61	10.61	10.61

Values are emergency department visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Run at 1 week ahead only.

Table S5: Mean Weighted Interval Score by Model and Horizon, 17 Areas, Columbus, Ohio, June 2024–May 2025

Model	1 wk	2 wk	4 wk
Graph neural network	4.06	5.03	6.31
Same model, map removed	4.70	5.37	6.39
LSTM	5.13	7.13	6.57
Cola−GNN	6.03	6.75	7.12
Gradient−boosted trees	6.96	5.47	6.59
Seasonal ARIMA	5.55	5.84	6.50
Seasonal blend, no learning	5.05	5.68	6.44
FluSight baseline	4.84	6.10	8.20
Persistence	4.67	5.89	8.45
Graph attention network[a]	7.11		
Dual−topology GCN[a]	8.71		
Climatology	6.75	6.75	6.75
Same week last year	5.75	5.75	5.75

Values are emergency department visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Run at 1 week ahead only.

Table S6: Mean Weighted Interval Score by Model, Season, and Horizon, 58 ZIP Codes, Chicago, Illinois, 2022–2026

	2022–2023			2023–2024			2024–2025			2025–2026		
Model	1	2	4	1	2	4	1	2	4	1	2	4
Graph neural network	1.16	1.31	1.48	1.08	1.14	1.23	1.09	1.13	1.26	1.00	1.07	1.17
Same model, map removed	1.27	1.38	1.56	1.14	1.20	1.26	1.16	1.22	1.31	1.08	1.15	1.20
LSTM	1.22	1.35	1.57	1.13	1.18	1.29	1.15	1.19	1.54	1.04	1.12	1.27
Cola-GNN	1.33	1.41	1.55	1.10	1.16	1.38	1.12	1.16	1.39	1.02	1.09	1.19
Gradient−boosted trees	1.29	1.41	1.51	1.14	1.20	1.27	1.21	1.27	1.37	1.11	1.15	1.20
Seasonal ARIMA	1.31	1.37	1.52	1.17	1.19	1.21	1.25	1.27	1.35	1.10	1.15	1.20
Seasonal blend, no learning	1.37	1.44	1.55	1.18	1.20	1.24	1.28	1.32	1.36	1.15	1.20	1.27
FluSight baseline	1.40	1.48	1.69	1.35	1.42	1.53	1.33	1.39	1.51	1.24	1.32	1.49
Persistence	1.44	1.52	1.68	1.33	1.39	1.48	1.35	1.42	1.52	1.25	1.32	1.50
Graph attention network[a]	1.49			1.16			1.12			1.07		
Dual-topology GCN[a]	1.59			1.43			1.18			1.11		
Climatology	1.69	1.69	1.69	1.28	1.28	1.28	1.51	1.51	1.51	1.26	1.27	1.27
Same week last year	1.76	1.76	1.76	1.63	1.63	1.63	1.57	1.57	1.57	1.47	1.47	1.47

Values are in Chicago Department of Public Health influenza-like illness activity levels (1–10), not rates, so they are not comparable in size with Boston or Columbus; lower is better. Column numbers are weeks ahead. No Chicago season was used in design, so all 4 are unseen. Learned models were tuned with the same budget as in Boston and Columbus.

[a] Run at 1 week ahead only.

Week-level comparisons

Each row compares the graph neural network with one model on the weeks of one season. The difference is graph neural network minus the other model, so a negative value favors the graph neural network. Intervals are 95% confidence intervals from a moving-block bootstrap over weeks (blocks of 4 weeks). *P* values are from the 2-sided Diebold–Mariano test with the Harvey–Leybourne–Newbold correction, Holm-adjusted across every model compared in that season and horizon (including the 1-week-only models and an untuned 10-seed LSTM, which are not shown). The same models enter the adjustment in all 3 cities.

Table S7: Week-Level Differences in Weighted Interval Score, Boston, Massachusetts, 2022–2025

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted *P*
2022–2023	1	Same model, map removed	−0.68 (−1.03, −0.36)	<.001
		LSTM	0.06 (−0.22, 0.34)	.62
		Cola-GNN	−0.45 (−1.32, 0.09)	.30
		Gradient-boosted trees	−0.73 (−1.15, −0.37)	<.001
		Seasonal ARIMA	−0.74 (−1.31, −0.37)	<.001
		Seasonal blend, no learning	−0.70 (−1.11, −0.39)	<.001
		FluSight baseline	−1.09 (−2.14, −0.47)	.001
		Persistence	−0.95 (−1.94, −0.31)	.005
2022–2023	2	Same model, map removed	−0.13 (−0.87, 0.60)	>.99
		LSTM	0.03 (−0.78, 0.63)	>.99
		Cola-GNN	−2.67 (−7.23, 0.22)	.95
		Gradient-boosted trees	−0.39 (−0.60, −0.20)	.001
		Seasonal ARIMA	−0.32 (−1.24, 0.57)	>.99
		Seasonal blend, no learning	−0.25 (−1.30, 0.71)	>.99
		FluSight baseline	−0.85 (−3.37, 0.85)	>.99
		Persistence	−0.60 (−3.12, 1.40)	>.99
2022–2023	4	Same model, map removed	0.09 (−0.18, 0.43)	>.99
		LSTM	−0.34 (−3.55, 2.68)	>.99
		Cola-GNN	−0.02 (−2.83, 2.77)	>.99
		Gradient-boosted trees	−0.36 (−0.81, 0.00)	>.99
		Seasonal ARIMA	0.01 (−1.51, 1.81)	>.99
		Seasonal blend, no learning	−0.35 (−2.16, 1.20)	>.99
		FluSight baseline	−1.55 (−6.37, 1.79)	>.99
		Persistence	−1.43 (−6.56, 2.72)	>.99
2023–2024	1	Same model, map removed	−0.34 (−0.48, −0.17)	.009
		LSTM	−0.14 (−0.42, 0.05)	.43
		Cola-GNN	−0.09 (−0.26, 0.05)	.43
		Gradient-boosted trees	−0.57 (−0.80, −0.37)	<.001
		Seasonal ARIMA	−0.73 (−0.95, −0.50)	<.001
		Seasonal blend, no learning	−0.93 (−1.24, −0.66)	<.001
		FluSight baseline	−1.13 (−1.48, −0.84)	<.001
		Persistence	−1.15 (−1.46, −0.88)	<.001
2023–2024	2	Same model, map removed	−0.30 (−0.50, −0.11)	.01
		LSTM	−0.78 (−1.05, −0.51)	<.001
		Cola-GNN	−0.22 (−0.71, 0.18)	.46
		Gradient-boosted trees	−0.54 (−0.77, −0.34)	<.001
		Seasonal ARIMA	−0.68 (−0.98, −0.46)	<.001
		Seasonal blend, no learning	−1.01 (−1.49, −0.68)	<.001

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
		FluSight baseline	−1.45 (−2.07, −0.99)	<.001
		Persistence	−1.50 (−2.03, −1.06)	<.001
2023–2024	4	Same model, map removed	−0.45 (−0.75, −0.17)	.03
		LSTM	−1.31 (−2.28, −0.53)	.03
		Cola-GNN	−0.90 (−2.10, −0.02)	.10
		Gradient-boosted trees	−0.81 (−1.09, −0.54)	<.001
		Seasonal ARIMA	−0.38 (−0.62, −0.10)	.04
		Seasonal blend, no learning	−0.90 (−1.42, −0.49)	.004
		FluSight baseline	−2.15 (−3.22, −1.35)	<.001
		Persistence	−2.34 (−3.42, −1.47)	<.001
2024–2025	1	Same model, map removed	−0.65 (−1.25, −0.18)	.009
		LSTM	−0.06 (−0.41, 0.25)	>.99
		Cola-GNN	−0.10 (−0.43, 0.24)	>.99
		Gradient-boosted trees	−1.54 (−3.20, −0.51)	.01
		Seasonal ARIMA	−2.70 (−5.17, −1.10)	<.001
		Seasonal blend, no learning	−0.83 (−1.39, −0.41)	<.001
		FluSight baseline	−1.55 (−2.83, −0.67)	.01
		Persistence	−1.22 (−2.41, −0.35)	.04
2024–2025	2	Same model, map removed	−0.31 (−0.57, −0.09)	.08
		LSTM	−0.33 (−0.74, 0.07)	.25
		Cola-GNN	−0.54 (−1.31, 0.06)	.25
		Gradient-boosted trees	−1.63 (−3.53, −0.48)	.19
		Seasonal ARIMA	−2.59 (−4.62, −1.19)	.009
		Seasonal blend, no learning	−1.21 (−1.98, −0.60)	.002
		FluSight baseline	−2.78 (−5.81, −0.86)	.17
		Persistence	−2.31 (−5.12, −0.39)	.19
2024–2025	4	Same model, map removed	−1.01 (−1.82, −0.40)	.06
		LSTM	−1.15 (−2.13, −0.43)	.07
		Cola-GNN	−0.74 (−1.70, −0.02)	.17
		Gradient-boosted trees	−1.31 (−2.69, −0.35)	.17
		Seasonal ARIMA	−2.41 (−3.89, −1.18)	.02
		Seasonal blend, no learning	−2.78 (−4.71, −1.07)	.06
		FluSight baseline	−5.98 (−12.10, −2.04)	.17
		Persistence	−5.88 (−12.08, −1.81)	.17

Table S8: Week-Level Differences in Weighted Interval Score, Columbus, Ohio, 2024–2025

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
2024–2025	1	Same model, map removed	−0.64 (−1.58, −0.06)	.07
		LSTM	−1.07 (−2.81, −0.08)	.07
		Cola-GNN	−1.96 (−4.87, −0.23)	.05
		Gradient-boosted trees	−2.90 (−7.04, −0.26)	.05
		Seasonal ARIMA	−1.49 (−3.34, −0.30)	.02
		Seasonal blend, no learning	−0.98 (−2.52, −0.04)	.07
		FluSight baseline	−0.78 (−1.47, −0.07)	.05
		Persistence	−0.60 (−1.13, 0.16)	.07
2024–2025	2	Same model, map removed	−0.35 (−1.14, 0.22)	.63
		LSTM	−2.10 (−4.72, −0.33)	.21

		Cola-GNN	−1.73 (−3.91, −0.30)	.23
		Gradient-boosted trees	−0.44 (−1.29, 0.12)	.63
		Seasonal ARIMA	−0.81 (−2.10, 0.02)	.63
		Seasonal blend, no learning	−0.66 (−1.93, 0.14)	.63
		FluSight baseline	−1.07 (−2.64, 0.23)	.63
		Persistence	−0.87 (−1.90, 0.27)	.63
2024–2025	4	Same model, map removed	−0.08 (−0.39, 0.21)	.93
		LSTM	−0.26 (−0.52, −0.10)	.25
		Cola-GNN	−0.80 (−1.44, −0.35)	.10
		Gradient-boosted trees	−0.28 (−0.72, 0.09)	.89
		Seasonal ARIMA	−0.19 (−0.44, 0.07)	.88
		Seasonal blend, no learning	−0.13 (−0.41, 0.13)	.93
		FluSight baseline	−1.88 (−5.62, 1.05)	.93
		Persistence	−2.14 (−4.48, −0.18)	.40

Table S9: Week-Level Differences in Weighted Interval Score, Chicago, Illinois, 2022–2026 (Activity Levels)

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
2022–2023	1	Same model, map removed	−0.11 (−0.17, −0.05)	<.001
		LSTM	−0.06 (−0.16, 0.00)	.02
		Cola-GNN	−0.17 (−0.43, 0.01)	.02
		Gradient-boosted trees	−0.13 (−0.24, −0.04)	<.001
		Seasonal ARIMA	−0.15 (−0.20, −0.11)	<.001
		Seasonal blend, no learning	−0.21 (−0.39, −0.07)	<.001
		FluSight baseline	−0.24 (−0.32, −0.15)	<.001
		Persistence	−0.28 (−0.35, −0.20)	<.001
2022–2023	2	Same model, map removed	−0.06 (−0.12, −0.03)	.06
		LSTM	−0.03 (−0.18, 0.06)	>.99
		Cola-GNN	−0.09 (−0.27, 0.03)	.63
		Gradient-boosted trees	−0.10 (−0.22, −0.00)	.28
		Seasonal ARIMA	−0.05 (−0.12, 0.05)	.63
		Seasonal blend, no learning	−0.12 (−0.25, −0.03)	.09
		FluSight baseline	−0.17 (−0.32, 0.02)	.09
		Persistence	−0.20 (−0.35, −0.02)	.02
2022–2023	4	Same model, map removed	−0.08 (−0.17, −0.00)	.81
		LSTM	−0.08 (−0.30, 0.10)	>.99
		Cola-GNN	−0.06 (−0.36, 0.14)	>.99
		Gradient-boosted trees	−0.02 (−0.09, 0.05)	>.99
		Seasonal ARIMA	−0.04 (−0.07, 0.01)	.80
		Seasonal blend, no learning	−0.07 (−0.26, 0.10)	>.99
		FluSight baseline	−0.20 (−0.44, 0.05)	.81
		Persistence	−0.20 (−0.40, 0.05)	.81
2023–2024	1	Same model, map removed	−0.06 (−0.10, −0.02)	.002
		LSTM	−0.04 (−0.10, 0.00)	.19
		Cola-GNN	−0.02 (−0.05, 0.01)	.22
		Gradient-boosted trees	−0.06 (−0.09, −0.03)	<.001
		Seasonal ARIMA	−0.09 (−0.12, −0.06)	<.001
		Seasonal blend, no learning	−0.10 (−0.15, −0.05)	<.001
		FluSight baseline	−0.27 (−0.35, −0.18)	<.001

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
		Persistence	−0.25 (−0.31, −0.18)	<.001
2023–2024	2	Same model, map removed	−0.06 (−0.10, −0.02)	.02
		LSTM	−0.04 (−0.11, 0.01)	.30
		Cola-GNN	−0.02 (−0.07, 0.02)	.30
		Gradient-boosted trees	−0.06 (−0.13, −0.01)	.09
		Seasonal ARIMA	−0.05 (−0.09, −0.02)	.007
		Seasonal blend, no learning	−0.06 (−0.11, −0.02)	.02
		FluSight baseline	−0.28 (−0.38, −0.18)	<.001
		Persistence	−0.25 (−0.32, −0.17)	<.001
2023–2024	4	Same model, map removed	−0.03 (−0.06, −0.00)	.29
		LSTM	−0.06 (−0.08, −0.04)	<.001
		Cola-GNN	−0.15 (−0.30, −0.02)	.30
		Gradient-boosted trees	−0.04 (−0.09, −0.00)	.30
		Seasonal ARIMA	0.02 (−0.02, 0.07)	.75
		Seasonal blend, no learning	−0.01 (−0.04, 0.03)	.75
		FluSight baseline	−0.30 (−0.48, −0.14)	.01
		Persistence	−0.25 (−0.37, −0.14)	.001
2024–2025	1	Same model, map removed	−0.08 (−0.12, −0.04)	<.001
		LSTM	−0.06 (−0.12, −0.02)	.003
		Cola-GNN	−0.03 (−0.05, −0.02)	<.001
		Gradient-boosted trees	−0.13 (−0.22, −0.06)	<.001
		Seasonal ARIMA	−0.16 (−0.27, −0.08)	<.001
		Seasonal blend, no learning	−0.20 (−0.31, −0.12)	<.001
		FluSight baseline	−0.24 (−0.32, −0.16)	<.001
		Persistence	−0.27 (−0.33, −0.21)	<.001
2024–2025	2	Same model, map removed	−0.08 (−0.15, −0.03)	.01
		LSTM	−0.06 (−0.13, −0.00)	.03
		Cola-GNN	−0.03 (−0.05, −0.01)	.02
		Gradient-boosted trees	−0.14 (−0.27, −0.05)	.02
		Seasonal ARIMA	−0.13 (−0.27, −0.04)	.02
		Seasonal blend, no learning	−0.19 (−0.31, −0.09)	<.001
		FluSight baseline	−0.26 (−0.35, −0.17)	<.001
		Persistence	−0.28 (−0.35, −0.21)	<.001
2024–2025	4	Same model, map removed	−0.06 (−0.13, 0.01)	.29
		LSTM	−0.28 (−0.56, −0.07)	.14
		Cola-GNN	−0.14 (−0.18, −0.10)	<.001
		Gradient-boosted trees	−0.11 (−0.19, −0.05)	.02
		Seasonal ARIMA	−0.09 (−0.22, −0.00)	.29
		Seasonal blend, no learning	−0.11 (−0.16, −0.06)	.002
		FluSight baseline	−0.25 (−0.40, −0.11)	.01
		Persistence	−0.26 (−0.37, −0.14)	<.001
2025–2026	1	Same model, map removed	−0.08 (−0.15, −0.04)	.001
		LSTM	−0.04 (−0.06, −0.02)	<.001
		Cola-GNN	−0.02 (−0.03, −0.01)	.003
		Gradient-boosted trees	−0.11 (−0.15, −0.06)	<.001
		Seasonal ARIMA	−0.10 (−0.13, −0.07)	<.001
		Seasonal blend, no learning	−0.15 (−0.17, −0.13)	<.001
		FluSight baseline	−0.24 (−0.30, −0.17)	<.001
		Persistence	−0.25 (−0.30, −0.19)	<.001

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
2025–2026	2	Same model, map removed	−0.07 (−0.15, −0.02)	.06
		LSTM	−0.05 (−0.07, −0.03)	<.001
		Cola-GNN	−0.02 (−0.04, 0.00)	.08
		Gradient-boosted trees	−0.08 (−0.13, −0.03)	.02
		Seasonal ARIMA	−0.07 (−0.12, −0.04)	.004
		Seasonal blend, no learning	−0.12 (−0.16, −0.08)	<.001
		FluSight baseline	−0.24 (−0.31, −0.17)	<.001
		Persistence	−0.25 (−0.30, −0.19)	<.001
2025–2026	4	Same model, map removed	−0.02 (−0.09, 0.04)	.90
		LSTM	−0.09 (−0.13, −0.05)	<.001
		Cola-GNN	−0.02 (−0.05, 0.01)	.79
		Gradient-boosted trees	−0.03 (−0.08, 0.03)	.90
		Seasonal ARIMA	−0.02 (−0.08, 0.03)	.90
		Seasonal blend, no learning	−0.10 (−0.14, −0.05)	<.001
		FluSight baseline	−0.31 (−0.44, −0.20)	<.001
		Persistence	−0.32 (−0.42, −0.23)	<.001

CI, confidence interval.

Other accuracy measures

WIS is the primary measure because it scores the whole forecast, including its range. Tables S10–S12 give the point measures often reported in earlier work. In all 3 cities the graph neural network had the lowest mean absolute error and root mean squared error at every horizon among the models in Tables S10–S12. Correlation ranks the models differently at some horizons because it ignores whether forecasts are too high or too low overall.

Table S10: Point Forecast Accuracy by Model and Horizon, 14 Neighborhoods, Boston, Massachusetts, 2022–2025

	Mean Absolute Error			Root Mean Squared Error			Correlation[a]		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	7.85	9.63	11.93	12.88	16.99	21.99	0.84	0.76	0.61
Same model, map removed	8.76	10.10	12.57	14.44	17.56	23.13	0.80	0.73	0.57
LSTM	7.96	10.19	13.02	13.18	18.01	24.09	0.87	0.76	0.54
Cola−GNN	8.19	11.04	12.27	13.77	20.56	23.11	0.85	0.70	0.61
Gradient−boosted trees	9.21	10.82	12.99	15.56	19.23	23.73	0.80	0.73	0.57
Seasonal ARIMA	9.69	11.17	13.09	16.45	19.38	23.40	0.78	0.70	0.59
Seasonal blend, no learning	9.06	10.88	13.69	15.07	19.03	25.33	0.79	0.69	0.49
FluSight baseline	9.50	11.89	16.23	15.31	20.21	29.37	0.78	0.68	0.44
Persistence	9.48	11.89	16.22	15.30	20.21	29.37	0.78	0.68	0.44

Errors are ED visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Pearson correlation between forecast and observed rates, computed for each area over the season and averaged across areas; higher is better.

Table S11: Point Forecast Accuracy by Model and Horizon, 17 Areas, Columbus, Ohio, June 2024–May 2025

	Mean Absolute Error			Root Mean Squared Error			Correlation[a]		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	6.02	7.32	8.62	10.17	13.21	16.82	0.83	0.74	0.62
Same model, map removed	7.04	7.65	8.71	12.41	14.41	17.23	0.81	0.75	0.64
LSTM	6.94	8.60	8.78	12.63	16.36	16.85	0.81	0.71	0.57
Cola−GNN	7.99	8.55	9.03	15.40	16.29	17.24	0.77	0.70	0.64
Gradient−boosted trees	9.66	7.67	8.70	18.95	14.47	17.34	0.60	0.75	0.65
Seasonal ARIMA	7.69	8.00	8.85	14.35	15.05	17.04	0.72	0.68	0.53
Seasonal blend, no learning	7.66	8.37	9.10	13.73	15.77	17.62	0.80	0.74	0.64
FluSight baseline	7.28	8.80	11.42	10.98	14.59	20.19	0.79	0.69	0.47
Persistence	7.28	8.80	11.42	10.98	14.59	20.19	0.79	0.69	0.47

Errors are ED visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Pearson correlation between forecast and observed rates, computed for each area over the season and averaged across areas; higher is better.

Table S12: Point Forecast Accuracy by Model and Horizon, 58 ZIP Codes, Chicago, Illinois, 2022–2026

	Mean Absolute Error			Root Mean Squared Error			Correlation[a]		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	1.62	1.75	1.96	2.31	2.47	2.67	0.60	0.54	0.45
Same model, map removed	1.73	1.85	2.03	2.48	2.60	2.77	0.51	0.46	0.39
LSTM	1.75	1.90	2.15	2.38	2.51	2.94	0.62	0.57	0.40
Cola-GNN	1.77	1.90	2.19	2.38	2.51	2.86	0.58	0.53	0.33
Gradient−boosted trees	1.81	1.93	2.04	2.48	2.60	2.74	0.51	0.47	0.42
Seasonal ARIMA	1.83	1.90	2.03	2.53	2.60	2.72	0.52	0.50	0.44
Seasonal blend, no learning	1.86	1.92	2.04	2.62	2.70	2.84	0.50	0.47	0.41
FluSight baseline	1.87	1.99	2.24	2.98	3.13	3.39	0.47	0.42	0.33
Persistence	1.87	1.99	2.24	2.98	3.13	3.39	0.47	0.42	0.33

Errors are in influenza-like illness activity levels (1–10); lower is better. wk, weeks ahead.
[a] Pearson correlation between forecast and observed rates, computed for each area over the season and averaged across areas; higher is better.

Severity alerts

Thresholds. For each neighborhood (or Columbus area), intensity thresholds come from the Moving Epidemic Method,[7] reimplemented in Python following the defaults of the `mem` R package, at the 50th, 90th and 98th percentile levels CDC uses. Each test season uses only earlier, non-pandemic seasons, and data stop the day the test season starts: Boston 2022–2023 uses 2017–2018 and 2018–2019; 2023–2024 adds 2022–2023; 2024–2025 adds 2023–2024. Columbus uses 2022–2023 and 2023–2024. The 2019–2020 through 2021–2022 seasons are left out because COVID-19 truncated or suppressed them. Chicago is not included: its measure is already the health department's graded 1–10 activity level.

Scores. A week is an event when the observed rate is at or above the threshold. The probability score uses each forecast's probability of being at or above the threshold, read from its calibrated 95% interval as a normal distribution; the Brier skill score compares its mean squared error with always

forecasting the pooled share of event weeks. The yes/no score alerts when the best-guess forecast is at or above the threshold; the Peirce skill score is the share of event weeks alerted minus the share of non-event weeks alerted. Both pool all neighborhood-weeks of a city's unseen seasons. Intervals for differences use the moving-block bootstrap of the main text. Code: `Code/compare_severity.py` and `Paper/scripts/make_severity.py`.

Table S13: Skill at Warning of Moderate and High Influenza Weeks, by Model, City, and Horizon, Boston, Massachusetts, and Columbus, Ohio

	Brier Skill Score (probability)						Peirce Skill Score (yes/no)					
	Moderate+			High+			Moderate+			High+		
Model	1	2	4	1	2	4	1	2	4	1	2	4
Boston, Massachusetts, 2022–2025 (3 seasons): 232 moderate and 82 high neighborhood-weeks of 2,103												
Graph neural network	0.61	0.41	0.11	0.40	0.12	−0.07	0.63	0.48	0.07	0.40	0.09	0.00
Same model, map removed	0.54	0.39	0.09	0.35	0.10	−0.07	0.62	0.45	0.06	0.32	0.10	0.00
LSTM	0.59	0.41	0.04	0.40	0.09	−0.05	0.63	0.48	0.06	0.33	0.01	0.00
Cola-GNN	0.57	0.23	0.11	0.39	0.03	−0.03	0.60	0.21	0.07	0.26	0.00	0.00
FluSight baseline	0.50	0.28	−0.16	0.32	−0.12	−0.59	0.67	0.53	0.26	0.59	0.29	0.00
Persistence	0.51	0.31	−0.09	0.37	−0.03	−0.51	0.67	0.53	0.26	0.59	0.29	0.00
Columbus, Ohio, 2024–2025: 248 moderate and 107 high neighborhood-weeks of 884												
Graph neural network	0.55	0.51	0.33	0.52	0.30	−0.04	0.68	0.65	0.35	0.51	0.36	0.02
Same model, map removed	0.51	0.48	0.31	0.38	0.20	−0.06	0.61	0.54	0.24	0.34	0.13	0.00
LSTM	0.54	0.36	0.29	0.36	−0.01	−0.04	0.66	0.32	0.42	0.16	0.00	0.02
Cola-GNN	0.46	0.37	0.30	0.02	−0.01	−0.08	0.49	0.34	0.13	0.00	0.00	0.00
FluSight baseline	0.42	0.36	0.14	0.42	0.22	−0.19	0.60	0.58	0.43	0.65	0.52	0.26
Persistence	0.43	0.37	0.17	0.44	0.26	−0.09	0.60	0.58	0.43	0.65	0.52	0.26

Column numbers are weeks ahead. Moderate+, at or above the 50th-percentile intensity threshold; High+, at or above the 90th. Scores above 0 show skill; higher is better. LSTM, long short-term memory network.

Table S14: Weeks Late in First Crossing a Neighborhood's Moderate Threshold, Median Over Area-Seasons

	Boston			Columbus		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	1	2	6	3	7	9
Same model, map removed	1	2	7	7	10	11.5
LSTM	1	3	4	8	15	5
Cola-GNN	1	3	5	14	15	8
FluSight baseline	1	2	4	1	2	4
Persistence	1	2	4	1	2	4

Positive values mean the forecast first crossed the threshold after the observed rate did; persistence is late by exactly the horizon by construction. Area-seasons in which the forecast never crossed are not included. wk, weeks ahead.

What each component adds on the unseen seasons

Each version of the graph neural network with one component removed (main text, Methods) was tuned with the same budget as the full model and averaged over 10 training runs; the task lists and commands are in the released code (`Code/results/_components/README.md`). Table S15

compares each version with the full model on the weeks of one season. "More error" is the version's mean WIS minus the full model's, as a percentage of the full model's, so a positive value means the component lowered error; intervals are 95% confidence intervals from a moving-block bootstrap over weeks (blocks of 4 weeks) on the same scale. *P* values are from the 2-sided Diebold–Mariano test with the Harvey–Leybourne–Newbold correction, Holm-adjusted across the 4 versions within each season and horizon.

Table S16 asks whether the version given only past rates forecasts like the comparison models that use only an area's own history. Here the reference is that version, and a positive value means the comparison model had more error. The 5 comparisons in each season and horizon form their own Holm family. Our criterion, set before the results were seen, was relative WIS within 5% of the seasonal blend's and no clear difference from it in most comparisons. Neither part was met: relative WIS was 5.2%, 7.1%, and 9.0% lower than the seasonal blend's at 1, 2, and 4 weeks in Boston, 6.1%, 4.9%, and 1.4% lower in Columbus, and 5.7%, 4.8%, and 2.9% lower in Chicago; the difference was clear in 12 of 24 season and horizon comparisons, each time favoring the version with past rates only.

Data version. The Boston Public Health Commission later revised 5 weekly neighborhood values before June 2025 (3 in the 2022–2023 test season). Every Boston model compared in the main text, including these versions, used the file as it stood before the revision, so that all were scored against the same observations. The multicity models in Table S18 were trained after the revision; all 5 revised values fall in their training weeks, none in the 2024–2025 season on which they were scored. The released code documents both versions.

Table S15: Change in Weighted Interval Score When One Component Is Removed From the Graph Neural Network, by City, Season, and Horizon, Boston, Massachusetts (2022–2025); Columbus, Ohio (2024–2025); and Chicago, Illinois (2022–2026)

City and Season	Weeks Ahead	Version	More Error, % (95% CI)	Adjusted *P*
Boston 2022–2023	1	No environment	0.8 (−2.2, 5.3)	.63
		No demographics	4.2 (0.8, 8.6)	.01
		No map	12.2 (6.5, 18.7)	<.001
		None of the three	9.4 (3.9, 15.2)	<.001
Boston 2022–2023	2	No environment	−0.3 (−10.4, 8.9)	>.99
		No demographics	2.3 (−9.0, 13.2)	>.99
		No map	1.7 (−8.2, 10.6)	>.99
		None of the three	1.4 (−7.7, 9.6)	>.99
Boston 2022–2023	4	No environment	−0.7 (−2.3, 0.8)	>.99
		No demographics	−1.4 (−4.0, 0.5)	.98
		No map	−0.8 (−3.9, 1.6)	>.99
		None of the three	0.7 (−3.0, 4.3)	>.99
Boston 2023–2024	1	No environment	−0.7 (−3.3, 2.7)	>.99
		No demographics	0.2 (−0.6, 1.3)	>.99
		No map	7.2 (3.6, 10.3)	.006
		None of the three	8.9 (4.7, 12.2)	.002
Boston 2023–2024	2	No environment	−1.5 (−6.5, 4.8)	.61
		No demographics	−1.0 (−2.4, 0.1)	.27
		No map	6.0 (2.3, 9.8)	.02
		None of the three	7.4 (1.8, 12.8)	.03
Boston 2023–2024	4	No environment	1.2 (−2.1, 4.6)	.51

City and Season	Weeks Ahead	Version	More Error, % (95% CI)	Adjusted P
		No demographics	1.2 (−0.2, 2.6)	.23
		No map	8.1 (3.1, 13.6)	.02
		None of the three	8.6 (3.6, 14.2)	.02
Boston 2024–2025	1	No environment	0.0 (−0.8, 0.8)	.90
		No demographics	1.1 (−0.3, 2.7)	.17
		No map	12.6 (3.4, 24.1)	.005
		None of the three	11.4 (2.9, 21.8)	.005
Boston 2024–2025	2	No environment	−1.1 (−6.1, 3.8)	.69
		No demographics	−4.5 (−15.1, 1.4)	.48
		No map	4.7 (1.4, 8.5)	.04
		None of the three	6.6 (2.0, 12.0)	.04
Boston 2024–2025	4	No environment	6.5 (1.2, 13.8)	.12
		No demographics	1.9 (−0.4, 5.5)	.28
		No map	12.4 (4.9, 22.1)	.03
		None of the three	11.1 (3.1, 21.5)	.07
Columbus 2024–2025	1	No environment	0.9 (−2.1, 5.3)	.53
		No demographics	−1.3 (−3.1, 0.0)	.35
		No map	15.8 (1.5, 40.1)	.09
		None of the three	16.6 (−0.2, 44.6)	.09
Columbus 2024–2025	2	No environment	3.7 (0.1, 9.1)	.22
		No demographics	2.1 (−1.3, 7.0)	.70
		No map	6.9 (−4.4, 22.6)	.70
		None of the three	7.5 (−4.5, 24.7)	.70
Columbus 2024–2025	4	No environment	−0.3 (−2.2, 1.4)	>.99
		No demographics	0.9 (−1.8, 3.7)	>.99
		No map	1.2 (−3.3, 6.5)	>.99
		None of the three	0.7 (−3.4, 5.2)	>.99
Chicago 2022–2023	1	No environment	−0.9 (−1.9, 0.4)	.13
		No demographics	−0.1 (−0.4, 0.4)	.80
		No map	9.2 (4.6, 14.6)	<.001
		None of the three	11.3 (7.1, 16.0)	<.001
Chicago 2022–2023	2	No environment	1.0 (−0.1, 2.2)	.23
		No demographics	−0.3 (−2.6, 1.4)	.70
		No map	4.9 (2.1, 9.0)	.03
		None of the three	5.0 (1.2, 8.6)	.03
Chicago 2022–2023	4	No environment	5.4 (1.7, 10.2)	.10
		No demographics	5.8 (0.9, 12.1)	.21
		No map	5.3 (0.2, 11.7)	.21
		None of the three	1.6 (−0.8, 3.6)	.21
Chicago 2023–2024	1	No environment	0.0 (−1.5, 1.2)	>.99
		No demographics	0.4 (−0.7, 1.4)	>.99
		No map	5.2 (2.3, 9.3)	<.001
		None of the three	5.6 (3.1, 9.0)	<.001
Chicago 2023–2024	2	No environment	0.2 (−0.4, 1.0)	.71
		No demographics	−0.3 (−0.8, 0.2)	.71
		No map	5.0 (1.8, 9.2)	.01
		None of the three	3.5 (1.4, 6.1)	.01

City and Season	Weeks Ahead	Version	More Error, % (95% CI)	Adjusted *P*
Chicago 2023–2024	4	No environment	2.7 (0.9, 4.2)	.02
		No demographics	0.3 (−1.0, 1.7)	.67
		No map	2.3 (0.2, 4.6)	.15
		None of the three	1.5 (−0.6, 3.0)	.27
Chicago 2024–2025	1	No environment	−0.6 (−1.2, −0.1)	.03
		No demographics	0.3 (−0.1, 0.8)	.19
		No map	6.9 (3.9, 11.5)	<.001
		None of the three	8.1 (5.4, 11.9)	<.001
Chicago 2024–2025	2	No environment	−1.8 (−2.9, −0.9)	.002
		No demographics	1.5 (−0.8, 4.0)	.16
		No map	7.5 (2.9, 13.3)	.005
		None of the three	6.8 (2.9, 11.8)	.005
Chicago 2024–2025	4	No environment	−2.7 (−5.3, −0.9)	.08
		No demographics	2.2 (0.4, 4.4)	.10
		No map	4.5 (−0.6, 10.2)	.14
		None of the three	4.8 (2.3, 8.2)	.02
Chicago 2025–2026	1	No environment	−0.7 (−1.4, −0.1)	.02
		No demographics	−0.2 (−0.9, 0.5)	.55
		No map	8.1 (3.8, 15.2)	.002
		None of the three	8.0 (5.0, 12.4)	<.001
Chicago 2025–2026	2	No environment	−0.2 (−1.4, 1.3)	>.99
		No demographics	−0.6 (−2.7, 1.2)	>.99
		No map	6.9 (2.1, 13.9)	.09
		None of the three	6.6 (3.4, 10.4)	.02
Chicago 2025–2026	4	No environment	0.7 (−0.9, 2.8)	>.99
		No demographics	0.2 (−0.7, 1.1)	>.99
		No map	1.9 (−3.0, 7.3)	>.99
		None of the three	2.2 (−1.1, 5.0)	.75

Table S16: Forecast Error of History-Only Models and the LSTM Compared With the Graph Neural Network Given Only Past Rates, by City, Season, and Horizon, Boston, Massachusetts (2022–2025); Columbus, Ohio (2024–2025); and Chicago, Illinois (2022–2026)

City and Season	Weeks Ahead	Comparison Model	More Error, % (95% CI)	Adjusted *P*
Boston 2022–2023	1	Seasonal blend	3.0 (0.5, 5.2)	.01
		Seasonal ARIMA	3.6 (−1.0, 8.4)	.09
		Persistence	7.1 (−2.7, 17.9)	.09
		FluSight baseline	9.4 (2.2, 21.0)	.02
		LSTM	−9.5 (−15.8, −3.6)	.01
Boston 2022–2023	2	Seasonal blend	1.8 (−2.3, 7.0)	>.99
		Seasonal ARIMA	2.7 (−1.7, 7.1)	.90
		Persistence	6.3 (−12.8, 28.6)	>.99
		FluSight baseline	9.5 (−4.8, 31.8)	.95
		LSTM	−1.8 (−8.8, 9.1)	>.99
Boston 2022–2023	4	Seasonal blend	2.5 (−8.9, 16.3)	>.99
		Seasonal ARIMA	−0.7 (−14.0, 10.1)	>.99

City and Season	Weeks Ahead	Comparison Model	More Error, % (95% CI)	Adjusted P
		Persistence	12.3 (−24.2, 56.2)	>.99
		FluSight baseline	13.4 (−16.2, 53.4)	>.99
		LSTM	2.5 (−26.2, 34.6)	>.99
Boston 2023–2024	1	Seasonal blend	9.9 (6.6, 13.9)	<.001
		Seasonal ARIMA	6.0 (2.7, 9.5)	.003
		Persistence	14.4 (10.5, 19.1)	<.001
		FluSight baseline	13.9 (9.4, 19.6)	<.001
		LSTM	−5.4 (−9.9, 0.8)	.05
Boston 2023–2024	2	Seasonal blend	11.7 (4.8, 19.3)	.004
		Seasonal ARIMA	5.7 (0.0, 11.4)	.04
		Persistence	20.7 (11.7, 31.1)	<.001
		FluSight baseline	19.8 (10.6, 30.5)	<.001
		LSTM	7.5 (3.2, 11.6)	.01
Boston 2023–2024	4	Seasonal blend	7.1 (−3.8, 18.3)	.53
		Seasonal ARIMA	−1.6 (−10.5, 5.8)	.68
		Persistence	31.0 (13.0, 52.2)	.03
		FluSight baseline	27.9 (10.4, 47.8)	.04
		LSTM	13.9 (3.7, 26.9)	.10
Boston 2024–2025	1	Seasonal blend	4.2 (0.6, 7.2)	.005
		Seasonal ARIMA	36.8 (12.8, 74.2)	.002
		Persistence	11.0 (−7.3, 29.6)	.18
		FluSight baseline	16.8 (1.5, 37.0)	.09
		LSTM	−9.2 (−19.9, −2.3)	.09
Boston 2024–2025	2	Seasonal blend	10.9 (4.6, 18.2)	.002
		Seasonal ARIMA	30.4 (9.3, 61.2)	.04
		Persistence	26.5 (−0.9, 66.1)	.22
		FluSight baseline	33.1 (7.6, 75.4)	.17
		LSTM	−1.6 (−7.3, 4.1)	.73
Boston 2024–2025	4	Seasonal blend	20.5 (7.0, 36.6)	.09
		Seasonal ARIMA	16.5 (−0.1, 34.1)	.29
		Persistence	54.6 (11.7, 118.2)	.29
		FluSight baseline	55.8 (14.5, 118.8)	.29
		LSTM	2.6 (−0.6, 5.4)	.29
Columbus 2024–2025	1	Seasonal blend	6.5 (0.2, 16.4)	.05
		Seasonal ARIMA	17.1 (5.3, 35.8)	.002
		Persistence	−1.5 (−40.0, 20.1)	>.99
		FluSight baseline	2.2 (−33.1, 23.6)	>.99
		LSTM	8.3 (0.3, 22.4)	.19
Columbus 2024–2025	2	Seasonal blend	5.2 (−0.3, 14.3)	.32
		Seasonal ARIMA	8.1 (−0.3, 19.9)	.30
		Persistence	9.1 (−26.9, 35.3)	.89
		FluSight baseline	12.8 (−24.6, 46.0)	.89
		LSTM	31.9 (8.3, 65.5)	.05
Columbus 2024–2025	4	Seasonal blend	1.4 (0.4, 2.7)	.13
		Seasonal ARIMA	2.3 (−3.0, 7.9)	.93
		Persistence	33.0 (−0.5, 72.3)	.38
		FluSight baseline	28.9 (−20.7, 92.9)	.93
		LSTM	3.4 (−2.4, 9.9)	.93

City and Season	Weeks Ahead	Comparison Model	More Error, % (95% CI)	Adjusted P
Chicago 2022–2023	1	Seasonal blend	5.7 (−3.6, 18.2)	.15
		Seasonal ARIMA	1.2 (−1.9, 3.9)	.36
		Persistence	11.4 (4.4, 16.3)	<.001
		FluSight baseline	8.2 (0.4, 13.8)	<.001
		LSTM	−5.4 (−9.6, 0.3)	.04
Chicago 2022–2023	2	Seasonal blend	4.2 (−5.1, 15.3)	.92
		Seasonal ARIMA	−1.0 (−5.5, 2.0)	>.99
		Persistence	10.0 (−0.2, 18.1)	.04
		FluSight baseline	7.6 (−2.8, 15.7)	.19
		LSTM	−2.5 (−9.6, 8.2)	>.99
Chicago 2022–2023	4	Seasonal blend	3.0 (−6.1, 13.8)	>.99
		Seasonal ARIMA	1.0 (−2.3, 3.3)	>.99
		Persistence	11.4 (−6.6, 27.6)	>.99
		FluSight baseline	11.9 (−6.5, 30.1)	>.99
		LSTM	3.9 (−10.2, 20.6)	>.99
Chicago 2023–2024	1	Seasonal blend	3.0 (0.0, 5.9)	.02
		Seasonal ARIMA	2.2 (−0.3, 4.1)	.08
		Persistence	16.3 (10.9, 21.1)	<.001
		FluSight baseline	18.0 (10.9, 24.8)	<.001
		LSTM	−1.6 (−5.6, 3.0)	.40
Chicago 2023–2024	2	Seasonal blend	2.0 (−0.3, 4.2)	.19
		Seasonal ARIMA	1.2 (−1.0, 3.1)	.43
		Persistence	17.5 (11.0, 23.5)	<.001
		FluSight baseline	20.2 (11.5, 28.5)	<.001
		LSTM	0.2 (−3.9, 5.2)	.93
Chicago 2023–2024	4	Seasonal blend	−0.8 (−3.1, 1.1)	.50
		Seasonal ARIMA	−3.2 (−6.6, −0.6)	.10
		Persistence	18.3 (9.7, 27.5)	.002
		FluSight baseline	22.4 (10.2, 36.1)	.009
		LSTM	3.2 (2.1, 4.4)	<.001
Chicago 2024–2025	1	Seasonal blend	9.2 (4.3, 15.9)	<.001
		Seasonal ARIMA	5.9 (1.1, 12.2)	.004
		Persistence	15.2 (10.4, 20.3)	<.001
		FluSight baseline	12.8 (5.7, 20.3)	<.001
		LSTM	−2.2 (−5.0, 0.5)	.09
Chicago 2024–2025	2	Seasonal blend	9.1 (3.7, 16.9)	.004
		Seasonal ARIMA	4.8 (−0.5, 12.8)	.18
		Persistence	17.0 (10.2, 23.6)	<.001
		FluSight baseline	14.9 (5.7, 23.6)	<.001
		LSTM	−1.4 (−5.2, 2.9)	.44
Chicago 2024–2025	4	Seasonal blend	3.5 (1.4, 5.6)	.02
		Seasonal ARIMA	2.2 (−3.6, 10.3)	.55
		Persistence	15.2 (4.7, 24.2)	.02
		FluSight baseline	14.5 (1.6, 25.9)	.09
		LSTM	17.0 (2.3, 35.3)	.13
Chicago 2025–2026	1	Seasonal blend	6.3 (2.2, 9.0)	<.001
		Seasonal ARIMA	1.6 (−0.7, 4.0)	.08
		Persistence	15.7 (9.1, 21.0)	<.001

City and Season	Weeks Ahead	Comparison Model	More Error, % (95% CI)	Adjusted P
		FluSight baseline	14.6 (7.4, 20.8)	<.001
		LSTM	−3.8 (−9.1, −0.2)	.07
Chicago 2025–2026	2	Seasonal blend	4.5 (0.6, 7.1)	.01
		Seasonal ARIMA	0.2 (−3.1, 3.4)	.88
		Persistence	15.7 (10.3, 20.9)	<.001
		FluSight baseline	15.1 (8.7, 21.4)	<.001
		LSTM	−1.8 (−5.6, 0.8)	.69
Chicago 2025–2026	4	Seasonal blend	6.2 (3.5, 8.0)	<.001
		Seasonal ARIMA	−0.1 (−3.4, 3.1)	.93
		Persistence	25.0 (15.9, 34.6)	<.001
		FluSight baseline	24.1 (13.6, 35.9)	<.001
		LSTM	5.8 (3.1, 7.8)	<.001

Design experience on the development season

Before the test seasons were scored, we chose the model's structure on the development season (2025–2026 in Boston; models trained on weeks from July 2022). To decide what to keep, we changed one part of the model at a time, retrained it from scratch with 10 training runs, and asked whether the change made its forecasts rise and fall more closely with what happened in each neighborhood. The measure is the correlation between forecast and observed rates, averaged over the 14 neighborhoods; Table S17 gives each change minus the default, so a positive value means the changed model tracked the observed curve better. To judge how large a difference has to be before it means anything, we trained the default 3 times on different random starting points. Those 3 results differed by up to 0.003, 0.006, and 0.031 at 1, 2, and 4 weeks, so smaller differences are within run-to-run noise. The table also counts the neighborhoods in which the change raised the correlation. These results were used only to make design choices. They come from the season used to design the model and are not evidence of performance, which comes from the unseen seasons in the main text, where removing the map, weather, the demographic measures, or all 3 is tested week by week (Figure 4; Table S15). There, the map's effect was confirmed, the demographic measures again added nothing, and the small gain from weather seen here was not.

Removing all links between neighborhoods lowered the correlation by 0.075 at 1 week and 0.093 at 2 weeks, in every neighborhood, but made no difference beyond noise at 4 weeks. Which links were used mattered much less. Keeping borders only, or removing the correlation or demographic links, changed the correlation by 0.006 or less at 1 and 2 weeks, and so did keeping only the links the model learns for itself. Removing weather lowered the correlation at every horizon and in every neighborhood. The 8 demographic measures did not help as inputs: removing them changed the correlation by 0.003 or less at 1 and 2 weeks and raised it by 0.027 at 4 weeks, in all 14 neighborhoods. They remained in the evaluated model. The blended baseline mattered most at 4 weeks, where replacing it with the current week alone lowered the correlation by 0.096.

Of the inputs we tried adding, Boston's influenza wastewater series raised the correlation by 0.089 at 4 weeks, in all 14 neighborhoods, but lowered it by 0.017 at 1 week. The series begins in July 2024 and covers under half of the training period, so the 4-week gain is uncertain, and wastewater was not part of the evaluated model. Statewide vaccination uptake also helped at 4 weeks (0.041). Neighborhood COVID-19 and RSV rates with COVID-19 wastewater made forecasts worse at every horizon. Links weighted by public transit ridership, measured in a separate study, changed the correlation by −0.001, −0.007, and +0.015; nearly every transit link duplicated an

existing link. Finally, in an earlier, untuned run of our reimplementation of the dual-topology model of Luo et al.,[3] adding a node for the surrounding area raised the correlation from 0.412 to 0.417, compared with the 0.135 gain reported in the original study.

Table S17: Change in Forecast Correlation When One Part of the Graph Neural Network Is Removed, Changed, or Added, Development Season, Boston, Massachusetts, June 2025–May 2026

	Change in Correlation (Neighborhoods Improved)[a]		
Part of the Model	1 Week Ahead	2 Weeks Ahead	4 Weeks Ahead
Removed			
All links between neighborhoods (map and learned)	−0.075 (0)	−0.093 (0)	−0.009 (6)
Hand-built links, keeping learned links	+0.006 (11)	−0.006 (3)	+0.006 (9)
Learned links, keeping hand-built links	−0.011 (0)	−0.007 (4)	+0.016 (14)
Correlation and demographic links (borders only)	+0.001 (9)	−0.004 (2)	−0.009 (0)
Correlation links	0.000 (9)	−0.003 (3)	+0.028 (13)
Demographic links	+0.003 (13)	+0.002 (9)	−0.001 (4)
Demographic measures as inputs	0.000 (6)	−0.003 (4)	+0.027 (14)
Demographic measures and links	+0.001 (9)	−0.002 (5)	+0.008 (13)
Weather	−0.012 (0)	−0.031 (0)	−0.025 (0)
Flag for suppressed weeks	−0.002 (5)	+0.006 (12)	+0.002 (8)
All inputs except past rates	−0.012 (0)	−0.008 (2)	−0.023 (1)
Correlation term in the training loss	−0.001 (3)	−0.006 (1)	+0.007 (11)
Changed			
Baseline: current week only	−0.010 (0)	−0.019 (1)	−0.096 (0)
No baseline	0.000 (7)	−0.059 (2)	−0.029 (3)
Link types kept separate	0.000 (9)	−0.011 (0)	−0.004 (5)
Smaller model	−0.005 (3)	−0.007 (2)	−0.003 (6)
Added			
Influenza wastewater	−0.017 (3)	+0.013 (11)	+0.089 (14)
Statewide vaccination uptake	0.000 (7)	+0.008 (13)	+0.041 (14)
Citywide ILI share	−0.004 (4)	−0.006 (2)	+0.021 (14)
Calendar terms	−0.001 (6)	−0.001 (8)	+0.025 (13)
COVID-19 and RSV rates, COVID-19 wastewater	−0.031 (0)	−0.155 (0)	−0.159 (0)
Transit ridership links[b]	−0.001	−0.007	+0.015

Abbreviations: ILI, influenza-like illness; RSV, respiratory syncytial virus.
The default model's mean correlation was 0.893, 0.802, and 0.582 at 1, 2, and 4 weeks. Not evidence of performance: this season was used to design the model.
[a] Mean per-neighborhood correlation between forecast and observed rates for the changed model minus the default, each averaged over 10 training runs; positive values favor the change. In parentheses, the number of the 14 neighborhoods in which the change raised the correlation. Differences within 0.003, 0.006, and 0.031 at 1, 2, and 4 weeks are within run-to-run noise.
[b] Separate study with the same default; per-neighborhood counts are not available.

Training on several cities at once

Because Columbus is forecast about as well by a Boston-trained model as by its own, we asked whether training on more than 1 city helps further. We used the same model and settings as in Table 5 of the main text (no neighborhood-specific learned links, settings fixed in advance, 10 training runs of 120 passes) and trained it on several cities together, using only weeks before the 2024–2025 test season in each. Each city kept its own map, its own scaling (each area's series standardized by its own training-period mean and standard deviation), and its own validation weeks. Every training batch came from 1 city, each city supplied the same number of batches per pass so that

Chicago's 58 areas did not outweigh Boston's 14 or Columbus's 17, and the stopping point was chosen on the average of the 3 cities' validation errors. We trained 2 versions: one on all 3 cities, which includes the target city's own earlier weeks, and one on the other 2 cities only, which, like the transferred models of Table 5, has seen no data from the target. Each was applied to each city's 2024–2025 season as is and after fine-tuning on local data, and compared week by week with the model trained on the target city alone (Holm adjustment over the 4 multicity versions at each horizon).

No multicity model was clearly more accurate than the locally trained one (Table S18). In Columbus, every multicity version had lower mean WIS than the local model at every horizon, by 3% to 8%, but none of these differences was clear (adjusted P values .06 to .40). The model trained on Boston and Chicago, with no Columbus data, beat the Boston-only model of Table 5 at 1, 2, and 4 weeks (3.85 vs 3.90, 4.80 vs 5.08, and 5.97 vs 6.45), clearly at 2 weeks ($P = .006$) and nearly so at 4 weeks ($P = .06$). In Boston and Chicago, which have long histories, their own data were best. Without further training, the multicity models were worse than the local ones at every horizon in both cities, clearly in 3 of the 4 Chicago comparisons at 1 and 2 weeks; after fine-tuning they tied the local models in all 12 comparisons. Training on 2 cities instead of 1 did not help Boston (adjusted $P > .99$ against each single-city source) and helped Chicago only slightly, clearly at 1 week (1.14 vs 1.16, $P = .04$) but not at 2 or 4 weeks.

Table S18: Forecast Error When the Model Is Trained on Several Cities at Once, Boston, Massachusetts; Columbus, Ohio; and Chicago, Illinois, Unseen Season June 2024–May 2025

			Trained on All 3 Cities[b]		Trained on the Other 2 Cities[c]		
Target City	Weeks Ahead	Trained Locally	As Is	Fine-Tuned	As Is	Fine-Tuned	Persistence
Boston	1	4.88	5.13	4.85	5.33	4.80	6.36
	2	6.15	6.63	6.07	6.96	6.34	8.95
	4	8.25	9.16	8.39	10.11	8.12	14.07
Columbus	1	4.11	3.87	4.00	3.85	3.95	4.67
	2	5.18	4.81	4.99	4.80	4.96	5.89
	4	6.49	6.02	5.95	5.97	6.07	8.45
Chicago[d]	1	1.09	1.10	1.08	1.14[a]	1.08	1.35
	2	1.12	1.15[a]	1.13	1.23[a]	1.12	1.42
	4	1.25	1.28	1.24	1.30	1.24	1.52

Abbreviations: WIS, weighted interval score.

Values are mean WIS in the target city's units: ED visits for influenza-like illness per 100,000 residents in Boston and Columbus, and the health department's 1–10 activity level in Chicago. Compare columns within a row, not values across cities. "Trained locally" and "Persistence" are the same as in Table 5 of the main text.

[a] Clearly different from the locally trained model (Holm-adjusted Diebold–Mariano $P < .05$); all 3 marked values are worse.

[b] Includes the target city's own weeks before the test season.

[c] No data from the target city; comparable to the transferred models of Table 5.

[d] Each area's series is scaled by its own training-period mean and standard deviation, so models trained elsewhere forecast in standardized units that are converted back to levels.

Supplementary figures

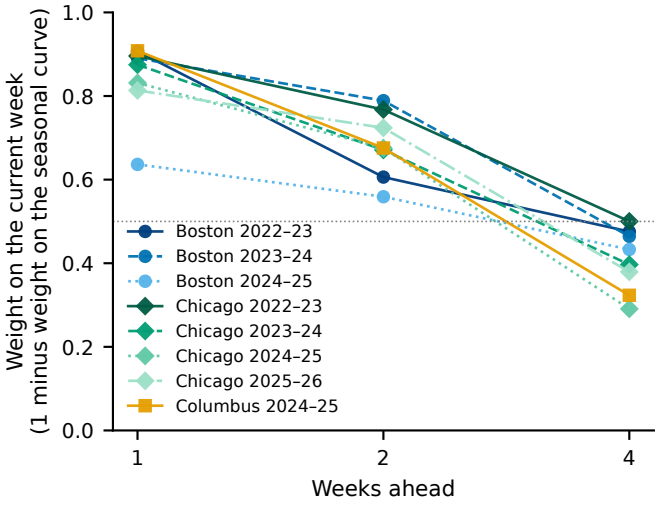


Figure S1. What the graph neural network leans on, by horizon, city, and unseen season. The model's baseline blends the current week's rate with the usual rate for that time of year; the y-axis is the learned weight on the current week (the rest goes to the seasonal curve). At 0.8 to 0.9 at 1 week in most seasons, the model mostly carries the latest value forward; by 4 weeks, it leans at least as much on the seasonal pattern. Boston, Massachusetts, 2022–2025; Columbus, Ohio, 2024–2025; Chicago, Illinois, 2022–2026.

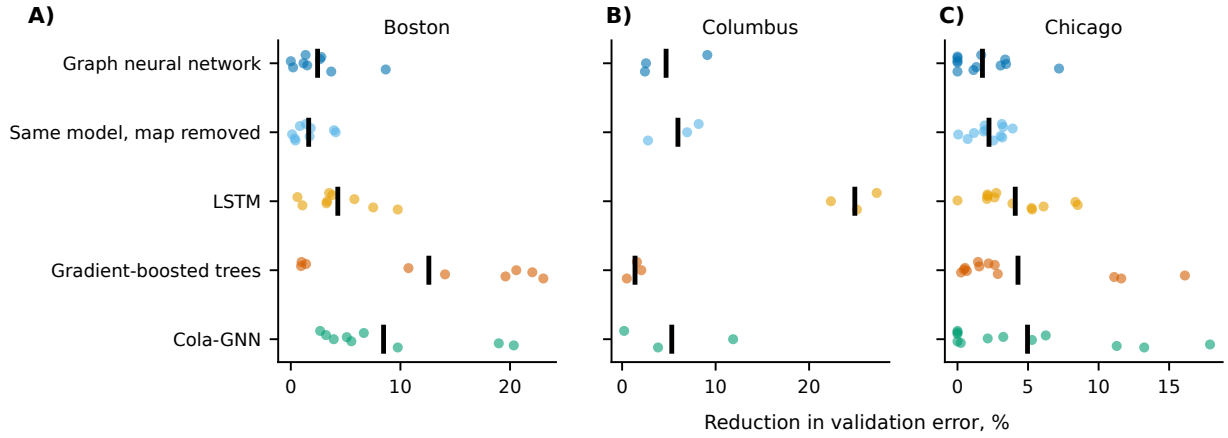


Figure S2. How much equal-budget tuning improved each model, by city: A) Boston, Massachusetts; B) Columbus, Ohio; C) Chicago, Illinois. Each dot is 1 test season and horizon; the percentage is the reduction in mean absolute error on validation weeks for the chosen settings compared with the model's default settings. Black bars are means. LSTM, long short-term memory network.

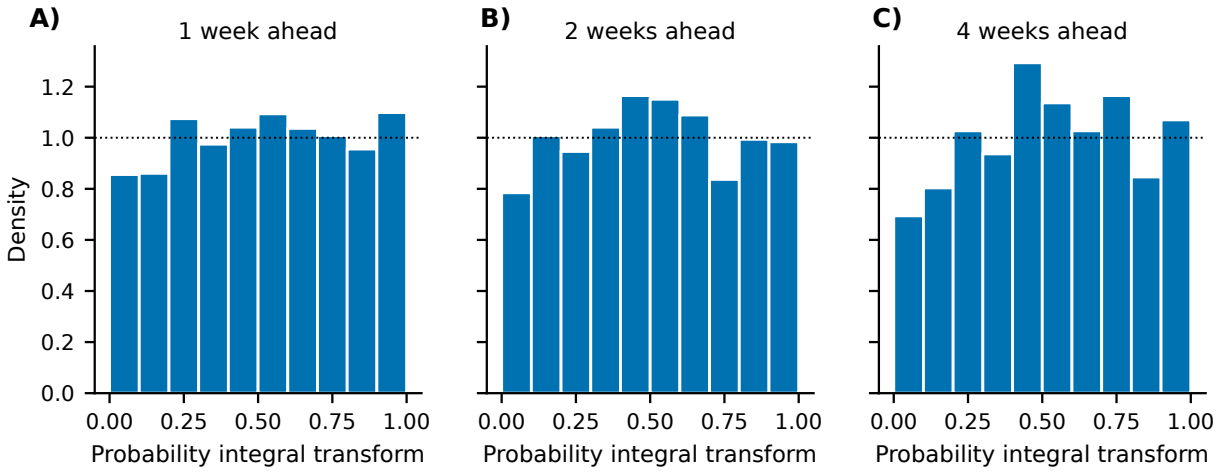


Figure S3. Calibration of the graph neural network's forecast distributions, 14 neighborhoods, Boston, Massachusetts, June 2022–May 2025: A) 1 week, B) 2 weeks, and C) 4 weeks ahead. Each histogram shows where observed values fell within the forecast distribution (the probability integral transform). A flat histogram at the dotted line means well-calibrated forecasts; a U shape would mean ranges that are too narrow, and a hump ranges that are too wide.

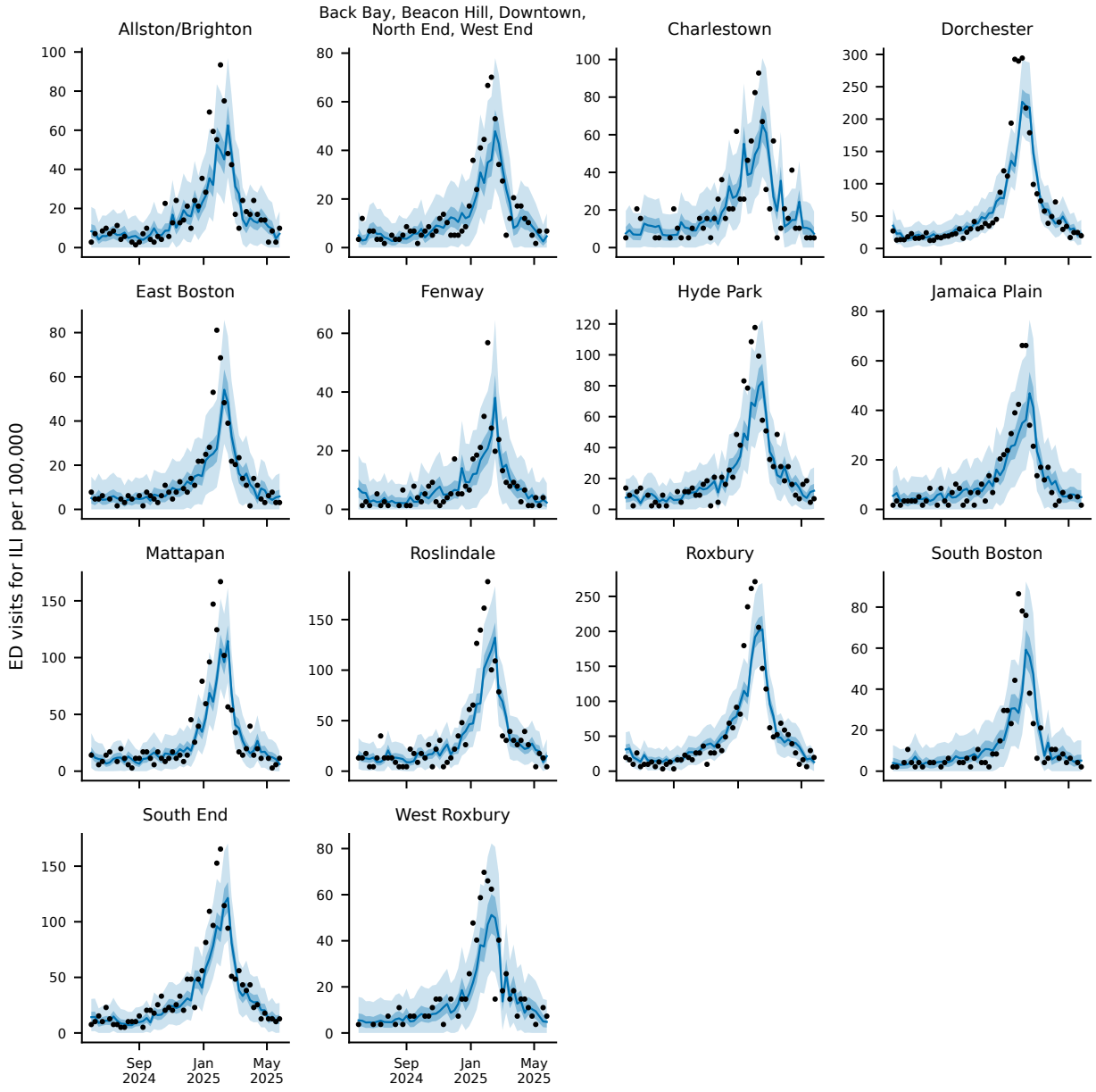


Figure S4. Forecasts made 2 weeks ahead for all 14 neighborhoods, Boston, Massachusetts, June 2024–May 2025. Lines show the graph neural network's median forecast, shading its 50% and 95% forecast intervals, and dots the observed emergency department visit rates for influenza-like illness (ILI) per 100,000 residents. Vertical scales differ by neighborhood.

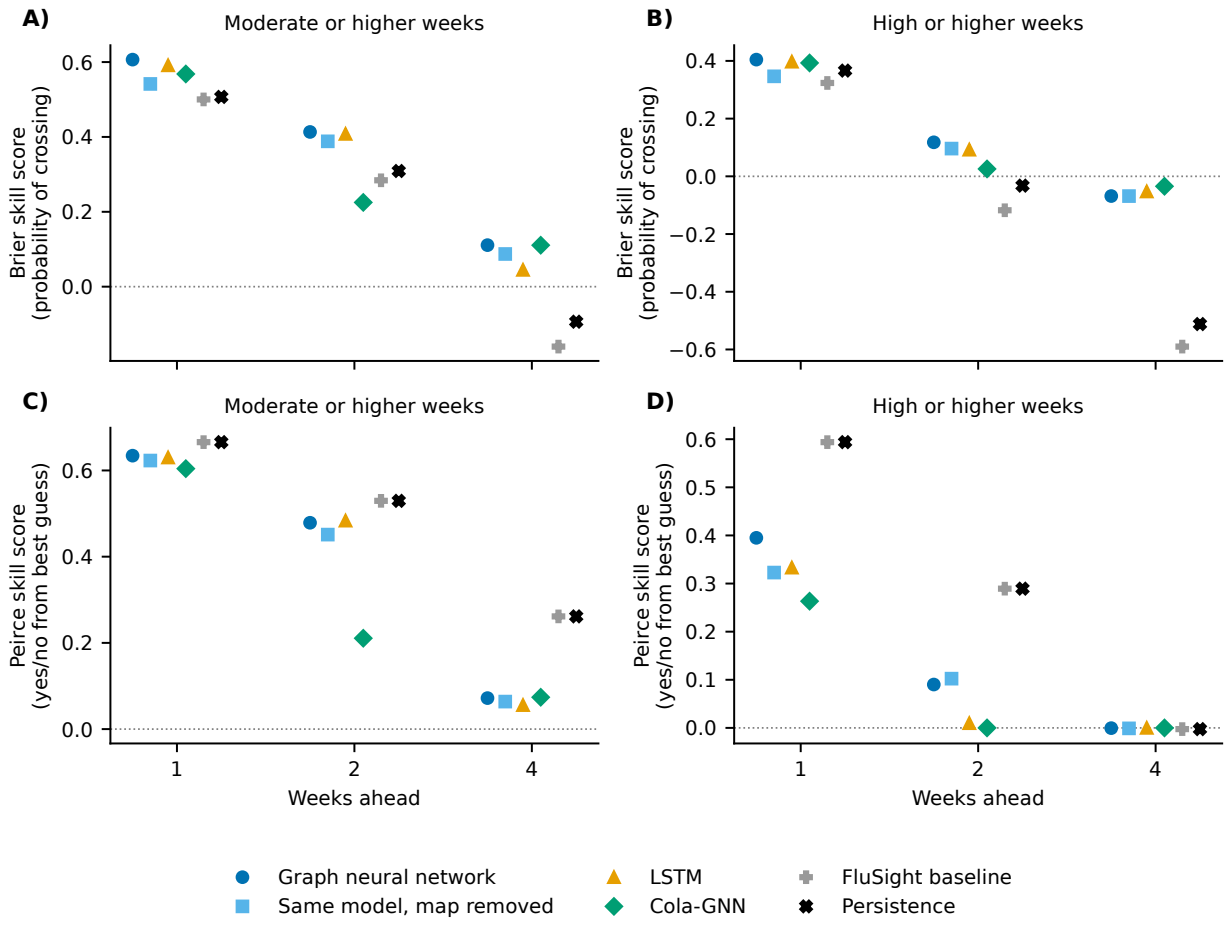


Figure S5. Skill at warning of moderate and high influenza weeks, 14 neighborhoods, Boston, Massachusetts, June 2022–May 2025. A, B) Brier skill score of the forecast's probability of being at or above the moderate (A) or high (B) threshold. C, D) Peirce skill score of a yes/no alert issued when the best-guess forecast is at or above the threshold. Values above the dotted line at 0 show skill. LSTM, long short-term memory network.

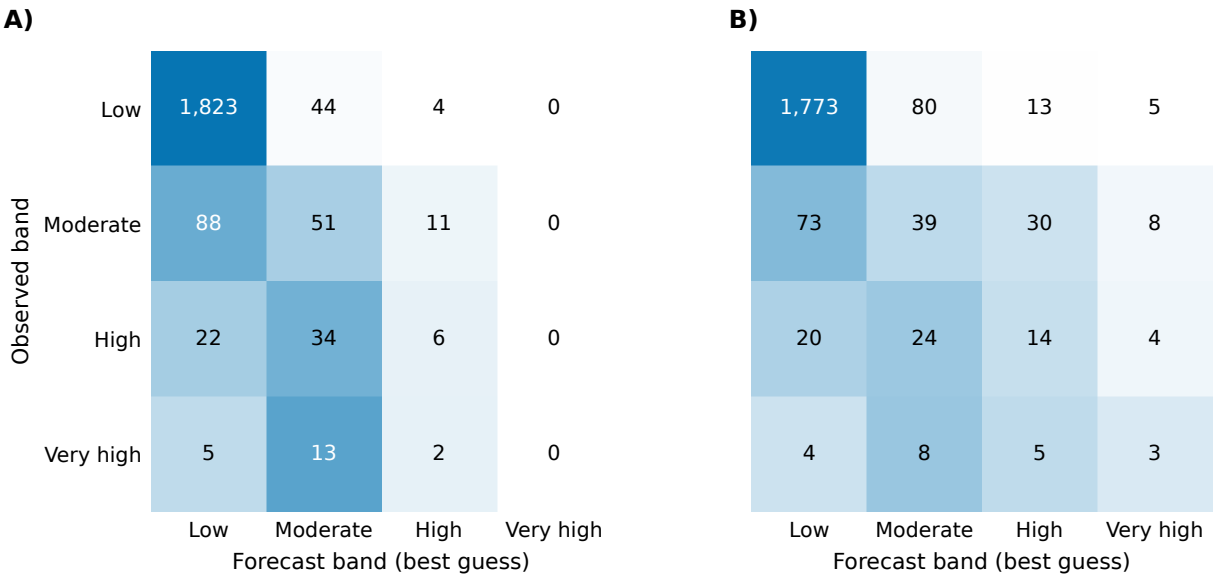


Figure S6. Severity band of each neighborhood-week, observed and forecast 2 weeks ahead, 14 neighborhoods, Boston, Massachusetts, June 2022–May 2025 (3 unseen seasons; 2,103 neighborhood-weeks). A) Graph neural network. B) Persistence (next week equals this week). Each forecast is placed in a band by its best guess (the median). Bands use each neighborhood's Moving Epidemic Method intensity thresholds, fitted only on seasons before each test season: low is below the 50th-percentile threshold, moderate from the 50th to below the 90th, high from the 90th to below the 98th, and very high at or above the 98th. Each cell gives the number of neighborhood-weeks with that observed band (row) and forecast band (column); shading shows the share of each observed band's weeks, so the diagonal shows how often the band was forecast correctly.

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